JAVA PRACTICAL PROGRAM REPORT

Prime Palindrome Number Checker

Program Name: PrimePalindrome.java

Language: Java

Topic: Prime number and palindrome checking

This report explains the supplied Java program, its logic, algorithm, class diagram, source code, execution commands, and sample console output.

1. Aim

To accept an integer and determine whether it is both a prime number and a palindrome number.

2. Requirements

• Java Development Kit (JDK)
• Text editor or Java IDE
• Save the source code as **PrimePalindrome.java**

3. Concepts and Logic

Prime check: A number less than or equal to 1 is not prime. The program checks divisors from 2 through n / 2; if any divisor divides n evenly, the prime flag becomes false.

Palindrome check: The program reverses the digits using remainder (%) and integer division (/). If the reversed number equals the original number, it is a palindrome.

Final condition: The number is reported as a prime palindrome only when both checks pass.

4. Algorithm

1. Start and read integer n using Scanner.

2. Set prime = true; if n is less than or equal to 1, set prime = false.

3. Check divisibility from i = 2 to n / 2; if divisible, set prime = false and stop the loop.

4. Copy n into temp and set reverse = 0.

5. Extract each last digit, append it to reverse, and remove the last digit from temp.

6. If prime is true and n equals reverse, print that it is a prime palindrome; otherwise print that it is not.

7. Close Scanner and end.

5. UML / Class Diagram

PrimePalindrome + main(args: String[]): void
Read integer n using Scanner
Check whether n is prime
Reverse digits to check palindrome
Combine both checks and print result
Close Scanner

The program has one class with a main method. It reads one integer, performs the prime and palindrome checks, prints a single result message, and closes the Scanner.

6. Source Code

```
import java.util.Scanner;

public class PrimePalindrome {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        // Check Prime
        boolean prime = true;

        if (n <= 1) {
            prime = false;
        }

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                prime = false;
                break;
            }
        }

        // Check Palindrome
        int temp = n;
        int reverse = 0;

        while (temp > 0) {
            int digit = temp % 10;
            reverse = reverse * 10 + digit;
            temp = temp / 10;
        }

        if (prime && n == reverse) {
            System.out.println(n + " is a Prime Palindrome Number");
        } else {
            System.out.println(n + " is not a Prime Palindrome Number");
        }

        sc.close();
    }
}
```

7. Compilation and Execution

```
javac PrimePalindrome.java
java PrimePalindrome
```

Sample input: 131. It is prime and its reverse is also 131.

8. Console Output

Captured by compiling and running the supplied program. The input is shown beside the prompt for readability.

```
Enter a number: 131
Enter a number: 131 is a Prime Palindrome Number
```

9. Result / Conclusion

The program checks both primality and palindrome property and reports whether the input satisfies both conditions. For sample input 131, the output confirms that it is a prime palindrome number.